

# Load Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q16-0579

Hard Rock 138 kV  
December 26, 2017

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0579. The Interconnection Customer proposes a 92.8 MW large load (data center) facility interconnecting at the Hard Rock 138 kV substation in the Virginia area. The request was received on February 06, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 212 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,305,000.

### **3. Network Impacts**

#### **3.1 Load Deliverability Analysis**

1. Facility: SEVEN MILE-ROSEMARY POWER STATION 138kV  
Contingency: MCCARTNEY 138 kV/1 kV Transformer Bank  
Loading: 118.3% (Rating: 245 MVA)  
DFAX: 30.9%  
Pre-project loading: 106.6%
2. Facility: CANON CITY-SEVEN MILE 138kV line  
Contingency: MCCARTNEY 138/1 kV Auto  
Loading: 115.9% (Rating: 242 MVA)  
DFAX: 30.9%  
Pre-project loading: 104.1%
3. Facility: MCCARTNEY 138 kV/1 kV Transformer Bank  
Contingency: MCCARTNEY-N TOWER TAP 345kV line  
Loading: 108.9% (Rating: 517 MVA)  
DFAX: 23.7%  
Pre-project loading: 104.6%
4. Facility: ROANOKE RAPIDS SUBSTATION-ROSEMARY POWER STATION 138kV  
Contingency: MCCARTNEY 138-1 kV Transformer  
Loading: 102.2% (Rating: 186 MVA)  
DFAX: 19.9%  
Pre-project loading: 92.3%

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                                                  | Required Upgrade                               | Cost Estimate | Allocation to Project |
|-----------------------------------------------------------|------------------------------------------------|---------------|-----------------------|
| SEVEN MILE<br>SUBSTATION-ROSEMARY POWER<br>STA 138kV Line | Rebuild line with higher-capacity<br>conductor | \$24,538,000  | 73% (\$18,005,000)    |
| CANON CITY-SEVEN MILE 138kV                               | Rebuild line with higher-capacity<br>conductor | \$6,227,000   | 73% (\$4,570,000)     |
| MCCARTNEY 138 kV/1 kV Transformer<br>Bank                 | Replace transformer with<br>higher-rated unit  | \$6,542,000   | 13% (\$824,000)       |
| ROANOKE RAPIDS-ROSEMARY<br>POWER STA 138kV                | Reconductor line                               | \$1,405,000   | 64% (\$906,000)       |
|                                                           | Total allocated to project                     |               | \$24,305,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.